

Advanced Bayesian Learning

Mixture models and Bayesian nonparametrics

Lecture 4

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Lecture overview

- Finite mixture models
- Dirichlet process mixtures
- MCMC for Dirichlet process mixtures

Finite mixture models

■ Mixture of normals

$$p(y) = \sum_{j=1}^k \pi_j \cdot \phi(y; \mu_j, \sigma_j^2)$$

■ **Allocation variables:** $l_i = j$ if y_i comes from $\phi(y; \mu_j, \sigma_j^2)$.

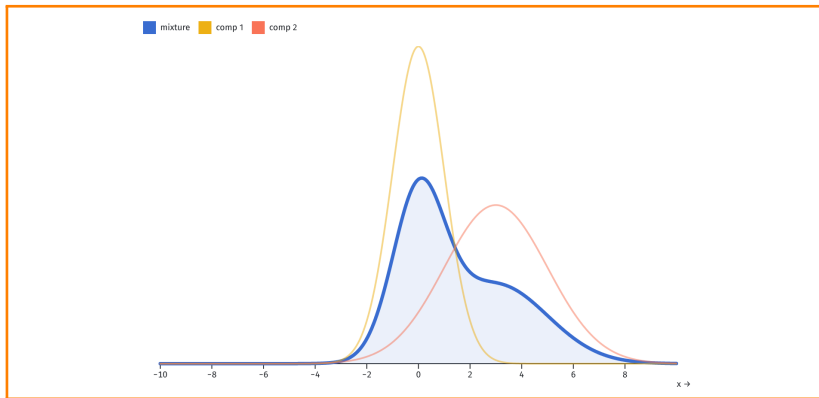
■ Let $l = (l_1, \dots, l_n)$ and $n_j = \sum_{i=1}^n (l_i = j)$.

■ **Gibbs sampling:**

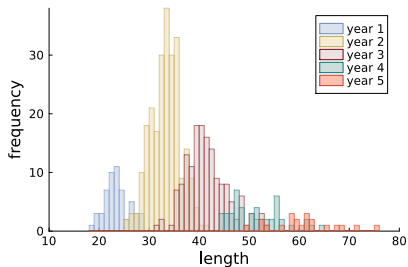
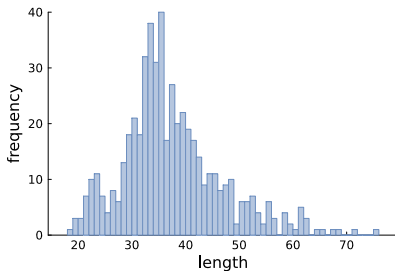
- ▶ $\pi_1, \dots, \pi_k \mid l, y \sim \text{Dirichlet}(a_1 + n_1, a_2 + n_2, \dots, a_k + n_k)$
- ▶ $\sigma_j^2 \mid l, y \sim \text{Inv-}\chi^2$ and $\mu_j \mid l, \sigma_j^2, y \sim \text{Normal}$ for $j = 1, \dots, k$.
- ▶ $l_i \mid \pi, \mu, \sigma^2, y \sim \text{Categorical}(\omega_{i1}, \dots, \omega_{ik}), i = 1, \dots, n,$

$$\omega_{ij} = \frac{\pi_j \cdot \phi(y_i; \mu_j, \sigma_j^2)}{\sum_{q=1}^k \pi_q \cdot \phi(y_i; \mu_q, \sigma_q^2)}.$$

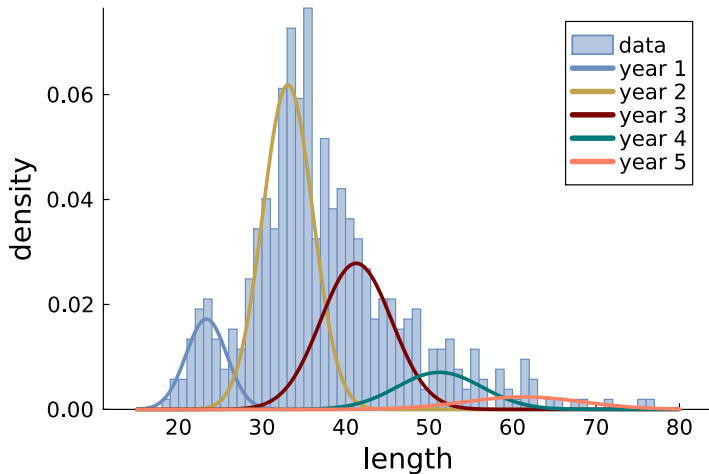
Interactive - Mixture of Normals



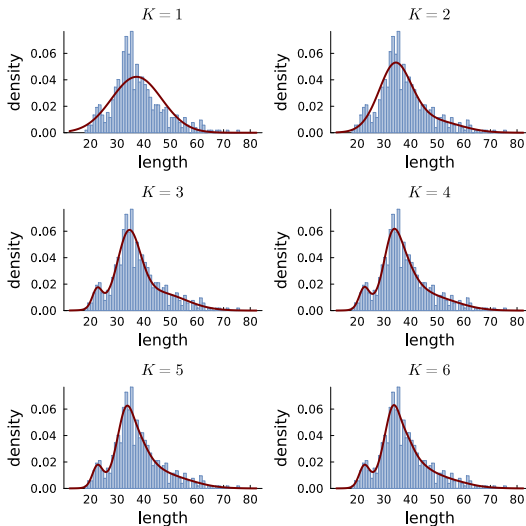
Fish length data with known yearly cohorts



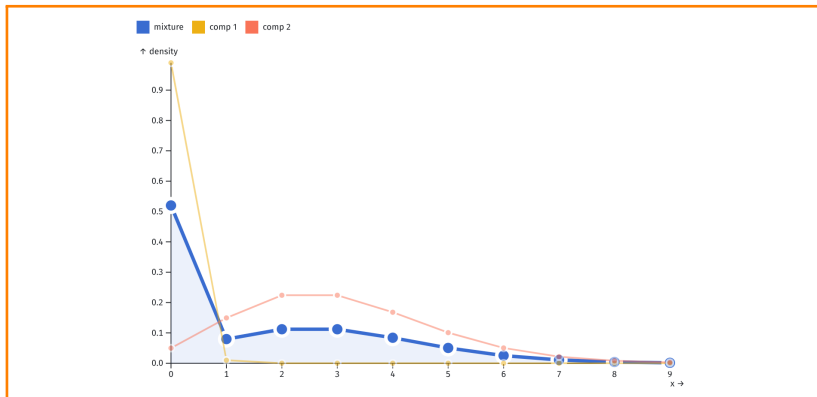
Fish length data - fit with known yearly cohorts



Fish length data - mixture of normals fit



Interactive - Mixture of Poissons



Infinite mixture models - DP mixtures

■ General mixture

$$f(y|P) = \int \mathcal{K}(y|\theta) dP(\theta)$$

where $\mathcal{K}(y|\theta)$ is a **kernel** and $P(\theta)$ is a **mixing measure**.

■ Student- t : $y \sim t_\nu(\mu, \sigma^2)$.

- ▶ $\mathcal{K}(y|\theta) = \phi(y|\mu, \lambda)$ where μ is fixed, $\theta = \lambda$ and
- ▶ $P(\theta)$ is the $Inv - \chi^2(\nu, \sigma^2)$ distribution.

■ Finite mixture of normals: $p(y) = \sum_{j=1}^k \pi_j \cdot \phi(y; \mu_j, \sigma_j^2)$.

- ▶ $\mathcal{K}(y|\theta) = \phi(y|\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$.
- ▶ $P(\theta)$ is discrete with $\Pr\{\theta = (\mu_j, \sigma_j^2)\} = \pi_j$, for $j = 1, \dots, k$.

■ Dirichlet Process Mixture: $P \sim DP(\alpha P_0)$, **infinite mixture**

$$f(y) = \sum_{h=1}^{\infty} \pi_h \mathcal{K}(y|\theta_h^*), \quad \pi \sim \text{Stick}(\alpha_0) \text{ and } \theta_h \stackrel{iid}{\sim} P_0$$

DP mixture is like a finite mixture with large k

- Infinite mixture: every observation has its own parameter

$$y_i \sim \mathcal{K}(\theta_i)$$

- DP is a.s. discrete $\Rightarrow$ ties: some θ_i exactly the same value.
- DP implies clustering of the θ_i
- Each observation has potentially its own parameter θ_i , but that parameter may be shared by other observations.
- Finite mixtures: observations share k parameter values.

$$y_i | I_i \sim \mathcal{K}(\theta_{I_i})$$

$$I_i | \pi \sim \text{Categorical}(\pi_1, \dots, \pi_k)$$

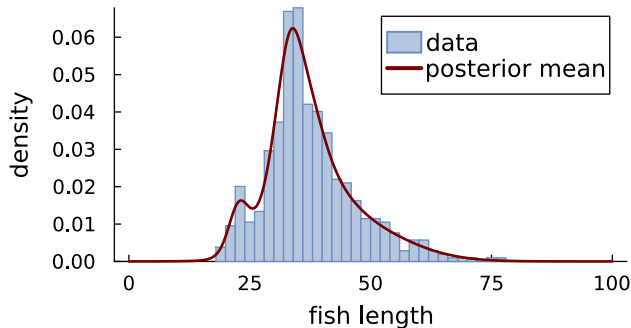
$$\theta_1, \dots, \theta_k \stackrel{iid}{\sim} P_0$$

$$\pi \sim \text{Dirichlet}(\alpha_0/k, \dots, \alpha_0/k)$$

- Finite mixture approaches DP mixture as $k \rightarrow \infty$ [Neal2000].

Dirichlet Process Mixture for fish length data

$$\alpha_0 = 1$$



$$\alpha_0 = 10$$



Marginalizing out P from a DP - Polya scheme

- Hierarchical representation of DP mixtures

$$y_i \sim \mathcal{K}(\theta_i), \quad \theta_i \sim P \quad P \sim DP(\alpha_0 P_0)$$

- We can marginalize out P to obtain the Polya scheme

$$p(\theta_i | \theta_1, \dots, \theta_{i-1}) \sim \left(\frac{\alpha_0}{\alpha_0 + i - 1} \right) P_0(\theta_i) + \left(\frac{1}{\alpha_0 + i - 1} \right) \sum_{j=1}^{i-1} \delta_{\theta_j}$$

- So $p(\theta_i | \theta_1, \dots, \theta_{i-1})$ is a mixture of the base measure P_0 and point masses at the previously “drawn” θ -values.
- ‘Marginalizing out P ’: integrate out π in the finite mixture model and let $k \rightarrow \infty$ [Neal2000].

DPs and the Chinese restaurant process

■ Polya scheme:

$$p(\theta_i | \theta_1, \dots, \theta_{i-1}) \sim \left(\frac{\alpha_0}{\alpha_0 + i - 1} \right) P_0(\theta_i) + \left(\frac{1}{\alpha_0 + i - 1} \right) \sum_{j=1}^{i-1} \delta_{\theta_j}$$

■ Chinese restaurant process:

- ▶ first customer sits at empty table and gets the dish θ_1^* from P_0 .
- ▶ second customer has options:
 - sit at first customer's table with probability $\frac{1}{1+\alpha_0}$ and eat θ_1^*
 - sit at a new table with probability $\frac{\alpha_0}{1+\alpha_0}$ and eat $\theta_2^* \sim P_0$.
 - $\vdots$
- ▶ the i th customer has options:
 - sit at table with dish θ_j^* with a probability proportional to n_j , the number of customers sitting at table j
 - sit at a new table with probability proportional to α_0 .

Gibbs sampling DP mixtures - marginalizing P

- Similar to Gibbs sampling for finite mixtures. Data augmentation with mixture component indicators l_i .
- 1 **Update component allocation** for i th observation y_i by sampling from multinomial

$$\Pr(l_i = j | \cdot) \propto \begin{cases} n_j^{(-i)} \mathcal{K}(y_i | \theta_j^*) & \text{for } j = 1, \dots, k^{(-i)} \\ \alpha \int \mathcal{K}(y_i | \theta) dP_0(\theta) & \text{for } j = k^{(-i)} + 1 \end{cases}.$$

- 2 **Update** the unique **parameter values** θ^* by sampling from

$$p(\theta_j^* | \cdot) \propto P_0(\theta_c^*) \prod_{i: l_i = j} \mathcal{K}(y_i | \theta_j^*)$$

- Note that, unlike finite mixtures, the l_i **are not independent** conditional on θ^* . This because we have marginalized out P . They have to be sampled **sequentially**.

Gibbs sampling for truncated DP mixtures

- Set upper bound N for the number of components.
Approximate DP mixture with $\pi_h = 0$ for $h = N + 1, \dots$
- Posterior sampling for infinite mixtures is now very similar to finite mixture. The I_i can be sampled independently.
- 1 Update component allocation for i th observation y_i by sampling from multinomial

$$\Pr(I_i = j | \cdot) \propto \pi_j \mathcal{K}(y_i | \theta_j^*) \quad \text{for } j = 1, 2, \dots, N.$$

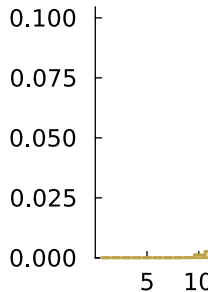
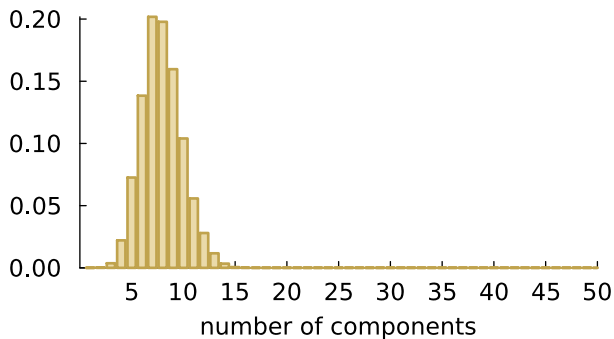
- 2 Update the stick-breaking weights $[\pi_h = V_h \prod_{\ell < h} (1 - V_\ell)]$

$$V_j | \cdot \sim \text{Beta} \left(1 + n_j, \alpha_0 + \sum_{q=j+1}^N n_q \right) \quad \text{for } j = 1, \dots, N-1.$$

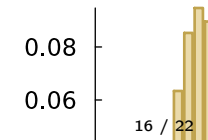
- 3 Update $\theta_1^*, \dots, \theta_N^*$ by sampling like in the finite mixture model.
Sample θ^* from prior $P_0(\theta)$ for empty clusters.

DPM for fish length data - number of comp

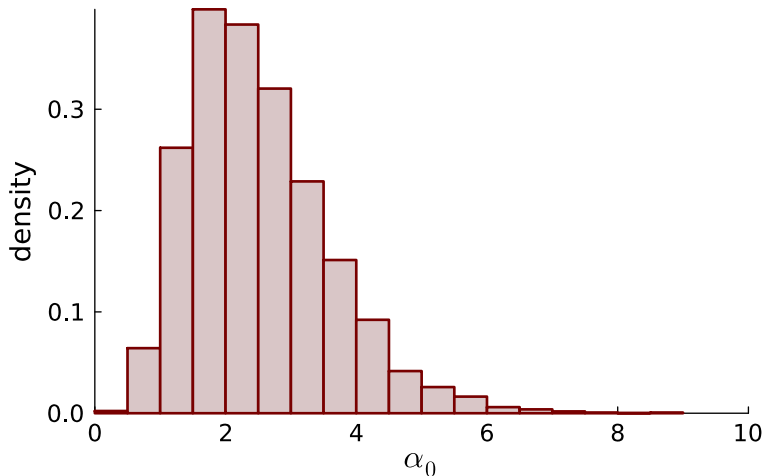
$$\alpha_0 = 1$$



$$\alpha_0 = 10$$



DPM for fish length data - posterior of α_0



MCMC for DP mixtures

- Let's look at the updating step:

$$\Pr(l_i = j | \cdot) \propto \begin{cases} n_j^{(-i)} \mathcal{K}(y_i | \theta_c^*) & \text{for } j = 1, \dots, k^{(-i)} \\ \alpha \int \mathcal{K}(y_i | \theta) dP_0(\theta) & \text{for } j = k^{(-i)} + 1 \end{cases}.$$

- A customer chooses table based on:

- ▶ the number of existing customers at the tables (with imaginary α_0 customers at a new table)
- ▶ how compatible the taste of the customer (y_i) is to the different dishes served at occupied tables (θ_c^*)
- ▶ how compatible the taste of the customer (y_i) is to the different dishes that *may* be served at a new table.
- ▶ A $P_0(\theta)$ with large variance is equivalent to an very experimental cook. You never know what you get ...

- α matters for the number of clusters (tables), but so does P_0 .
- α_0 can be learned from data. Just add updating step.
- P_0 may have hyperparameters (e.g. $P_0 = N(\mu, \sigma^2)$). Just add updating steps for those.

Mixture of multivariate regressions - Model

- The response vector y is p -dim. Covariates x is q -dim.
- The model is of the form

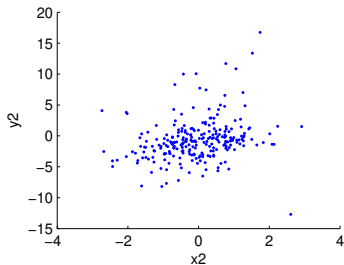
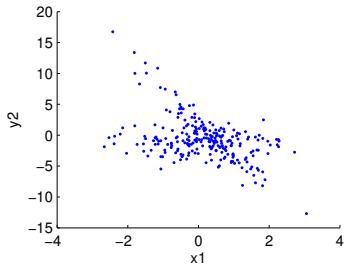
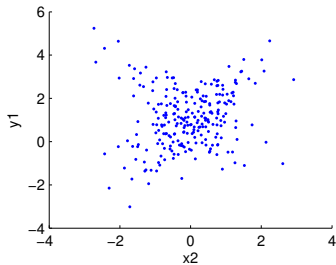
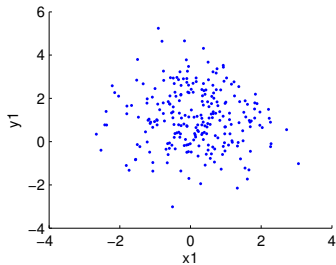
$$p(y|x) = \sum_{j=1}^{\infty} \pi_j \cdot N(y_i | B_j x_i, \Sigma_j)$$

- Each mixture component is a Gaussian multivariate regression with its own regression coefficient and covariance matrix:

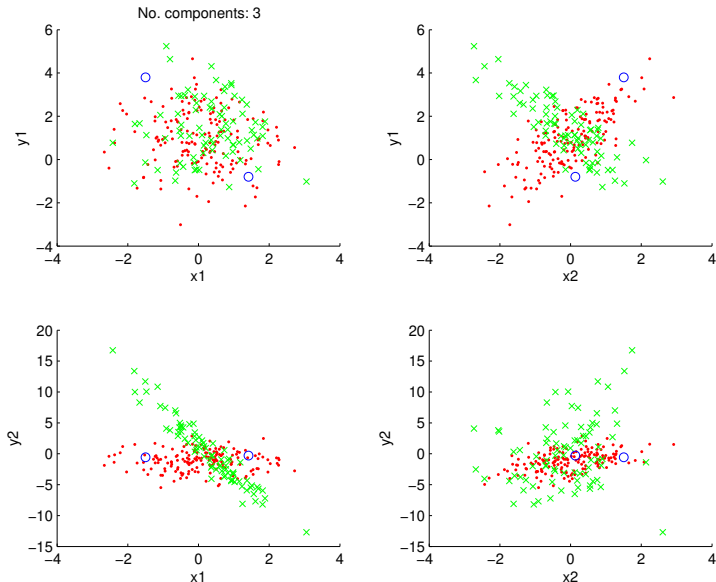
$$\underset{p \times 1}{y_i} = \underset{p \times q}{B_j} \underset{q \times 1}{x_i} + \underset{p \times 1}{\varepsilon_i}, \quad \varepsilon_i \stackrel{iid}{\sim} N(0, \Sigma_j)$$

- The mixture weights follow a DP stick prior $\pi \sim \text{Stick}(\alpha_0)$.

Mixture of multivariate regressions - Data



Mixture of multivariate regressions - DPM



Mixture of multivariate regressions - DPM

